

Language is the means for communication for humans. By studying language, we come to understand more about the world. If we can succeed at building computational mode of language, we will have a powerful tool for communicating around the world. We look at how we can exploit knowledge about the world, in combination with linguistic facts, to build computational natural language systems. Natural language processing (NLP) is the process of computer analysis of input provided in a human language (natural language), and conversion of this input into a useful form of representation.

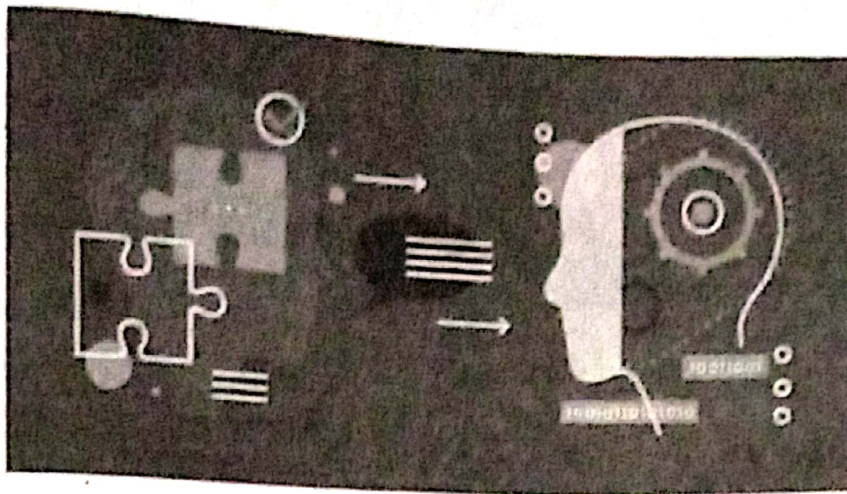


Figure 7.8: Natural language processing

NLP is one field of AI that processes or analyzes written or spoken language. It involves processing of speech, grammar and meaning. NLP is composed of two part: NLU (Natural Language Understanding) and NLG (Natural Language generation). Understanding of any language requires detailed knowledge of that language.

7.3.1 Components of NLP

There are two components of NLP:

1. Natural Language Understanding (NLU)

Natural language understanding (NLU) helps the machine to understand and analyse human language by extracting the metadata from content such as concepts, entities, keywords, emotion, relations, and semantic roles.

NLU mainly used in business applications to understand the customer's problem in both spoken and written language.

NLU involves the following tasks:

- It is used to map the given input into useful representation.
- It is used to analyze different aspects of the language.

2. Natural Language Generation (NLG)

Natural language generation (NLG) acts as a translator that converts the computerized data into natural language representation. It mainly involves Text planning, Sentence planning, and Text Realization.

7.3.2 Natural Language Processing Steps/Processes

The input of NLP system can be in the form of written text or speech. Quality of input decides the possible errors in language processing i.e., high quality input leads to correct language understanding and vice-versa. After that inputs are segmented in which text inputs are divided in segments (chunks) and the segments are analyzed. Each chunk is called a frame.

After that each segment is analyzed through various step as shown in figure below:

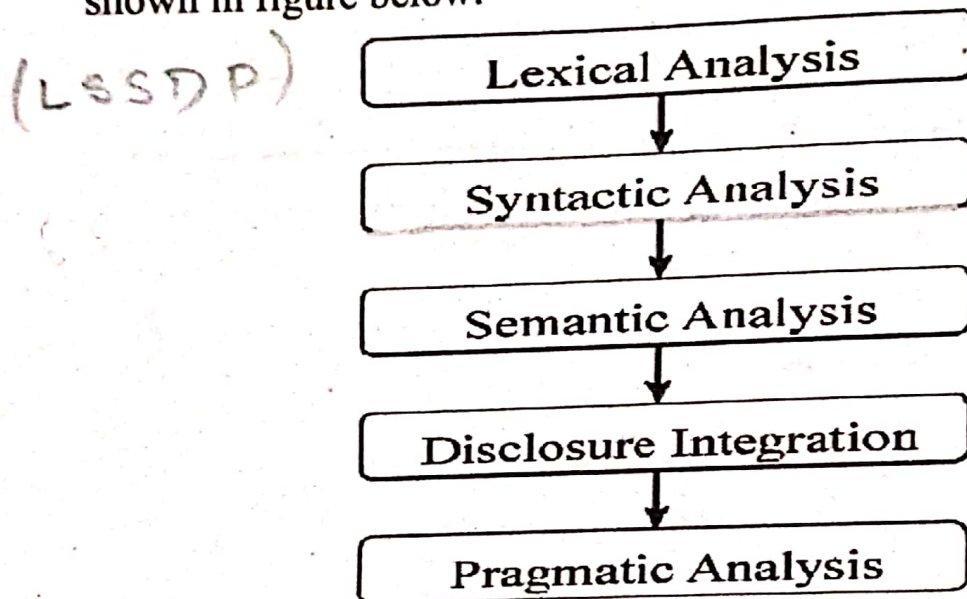
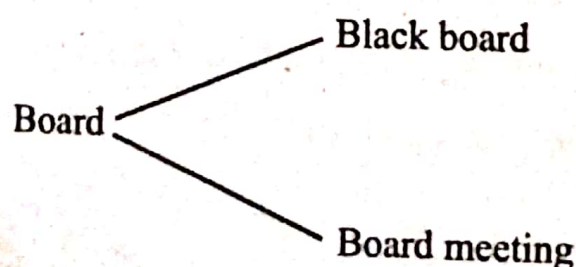


Figure 7.9: Steps in natural language processing

1. Lexical or Morphological Analysis

The lexicon of a language is its vocabulary that includes its words and expressions. Morphology is the identification, analysis and description of structure of words. The lexical analysis objective is to divide the text into paragraphs, sentences, words and analyze the structure of words. The lexical analysis cannot be performed in isolation from morphological and syntactic analysis.



For example board can have different meaning on different context; board might refers to the board in which we write or the board meeting(meeting of the board members of the organization).

2. Syntactic Analysis

Syntactic analysis takes an input sentence and produces a representation of its grammatical structure. A grammar describes the valid parts of speech of a language and how to combine them into phrases. The English grammar is nearly context free. A computer grammar specifies which sentences are in a language and their parse trees. A parse tree is a hierarchical structure that shows how the grammar applies to the input. Each level of the tree corresponds to the application of one grammar rule. The sentences such as "the school goes to boy" is rejected by English syntactic analyzer. For example,

The red bird pecks the grain.

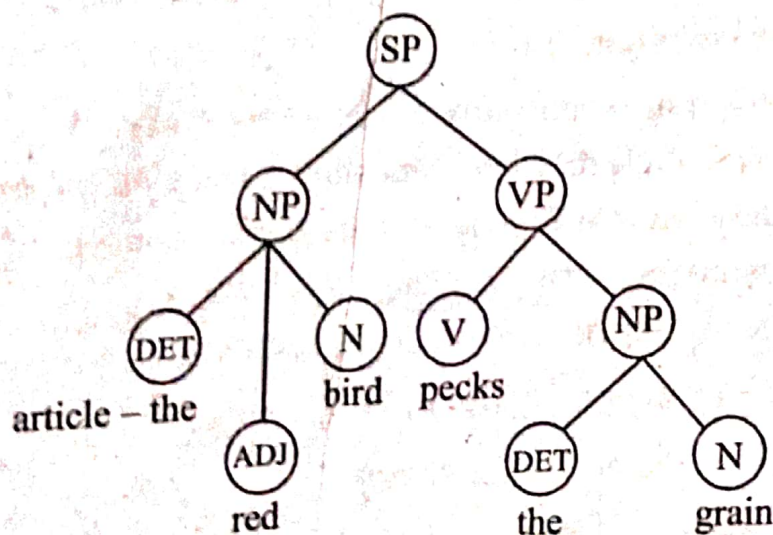


Figure 7.10: Parse tree

3. Semantic Analysis

Semantic analysis is a process of converting the syntactic representations into a meaningful representation. This analysis involves the following tasks:

- Word sense determination
- Sentence level analysis

Word sense:

Words have different meanings in different contexts.

E.g., Ram loves fish.

Here the meaning of loves can be 'likes' and also 'eats'.

E.g., Mary had a bat in her office.

Here, bat can be 'a baseball thing' or 'a flying mammal'.

Sentence level meaning

Once the words are understood, the sentence must be assigned some meaning.

E.g., Ram ate hot ice-cream.

Semantic analyzer rejects this sentence as it states hot ice-cream would make no sense.

E.g., Colorless blue idea.

This would be rejected by the analyzer as colorless blue do not make any sense together.

4. Discourse integration

The meaning of an individual sentence may depend on the sentences that precedes it and may influence the meaning of the sentences that follow it. It covers the concept of pronouns and referring expressions. The meaning of individual sentence is depends on previous sentence.

E.g., Bill had a red balloon. John wanted it.

Here, "it" means "a red ballon".

5. Pragmatic Analysis

Pragmatics comprises aspects of meaning that depend upon the context or upon facts about real world. These aspects include:

- Logical inferences, that can be drawn from the meanings of a set of propositions.

E.g., Close the window

Here, it should have been interpreted as a request rather than an order.

E.g., Jack fell. Jill brought him a band-aid.

Here, it should be understood as: Jack got hurt and Jill wanted to help.

✓ 7.3.3 Natural Language Processing AIDS or Tools

NLP can be understood or analyzed using some aids or tools.

1. ✓ **Lexicon or Dictionary**

A *lexicon* defines the words of a language that a system knows about. This includes common words and words that are specific to the domain of the application. Entries include meanings for each word and its syntactic and morphological behavior.

2. ✓ **Morphological Analysis System**

Morphological analysis is the process of recognizing the suffixes and prefixes that have been attached to a word. We do this by having a table of affixes and trying to match the input as: prefixes + root + suffixes.

For example: adjective +ly mean adverb.

“-s, -es” can be either a plural noun or a verb.

“-d, -ed” can be either a past tense or a perfect participle.

Following are some examples in which a word can have multiple form or changes its form.

E.g., Green can be transformed into greenness (adjective, noun)

E.g., Boat can be boats (singular, plural)

E.g., Bill slept. Bill's bed. (subjective case, possessive case)

E.g., I walk. I am walking. (present, progressive)

E.g., I walked. I will walk. I had been walking. (past, future, past progressive).

E.g., I walk. They walk. (first person singular, third person plural)

Problems in morphological analysis system:

- doubled consonants (sit, sitting)
- changing y to i (fly, flies)
- abbreviations (CAN)
- naming word (Cook)
- wrong spelled word (Hare instead hair)
- unrecognized words:
 - real words not in the dictionary
 - names, etc that cannot be listed (names, account nos., library call nos)

3. Syntactic Analysis System

It takes an input sentence and produces a representation of its grammatic structure. Grammar (mostly context free grammar) and parser are used.

4. Semantic Analysis System

It converts the syntactic representation into a meaning representation.

It involves:

- Word sense determination
- Sentence level analysis
- Knowledge Representation

5. Pragmatic Analysis System

In this system, we study how knowledge about the world and language conventions interacts with the literal meaning which depends upon context and facts about real world. Pragmatic analysis system includes:

- Pronouns and referring expressions.
- Logical interfaces that can be drawn from meaning of a set of propositions.

- Inference about what the producer of a sentence is trying to do.

7.3.4 Natural Language Analysis Techniques

There are two commonly used techniques in natural language processing. They are :

1. Keyword analysis or pattern matching

In this analysis, accept the given sentence as input then segment the sentence and identify keywords in each segment.

- If keyword is present
 - If only one keyword is present, give suitable reply as of keyword
 - If more keywords are present, prioritize them and give suitable reply as of keyword
- If no keyword present in the segment, give a random reply.

2. Syntactic driven parsing technique

In his technique, words can fit together in higher level units such as phrases, clauses and sentences so interpretations of larger groups of words are built up out of the interpretation of their syntactic constituent words or phrases. As interpretation of input is done as a whole it can be obtained by application of grammar that determines what sentences are legal in the language that is being parsed.

7.3.5 Natural Language Processing Problems

1. The same expression means different things in different context.

- Where's the water? (Chemistry lab? Must be pure)
- Where's the water? (Thirsty? Must be drinking water)
- Where's the water? (Leaky roof? It can be dirty)

2. No natural language program can be complete because of new words, expression, and meaning can be generated quite freely
 - I'll fax it to you
3. There are lots of ways to say the same thing.
 - Ram was born on October 11.
 - Ram's birthday is October 11.
4. Sentence and phrases might have hidden meanings
 - Out of sight, out of mind means invisible, idiot respectively.
5. Problem due to syntax and semantics
6. Problem due to extensive use of pronouns (semantic issue)
 - E.g., Ravi went to the supermarket. He found his favorite brand of coffee in rack. He paid for it and left.
 - It denotes??
7. Use of grammatically incorrect sentence
 - He rice eats. (syntax issue)
8. Use of conjunctions to avoid repetition of phrases cause problem in NLP
 - Ram and hari went to restaurant. Ram had a cup of coffee while hari had tea.

7.3.6 Natural Language Processing Applications

Following are the applications of NLP:

(QSSMSS CI)

i. Question answering

Question answering focuses on building systems that automatically answer the questions asked by humans in a natural language.

ii. Spam detection

Spam detection is used to detect unwanted e-mails getting to a user's inbox.

iii. Sentiment analysis

Sentiment analysis is also known as *opinion mining*. It is used on the web to analyse the attitude, behaviour, and emotional state of the sender. This application is implemented through a combination of NLP (natural language processing) and statistics by assigning the values to the text (positive, negative, or natural), identify the mood of the context (happy, sad, angry, etc.)

iv. Machine translation

Machine translation is used to translate text or speech from one natural language to another natural language.

Example: Google translator

v. Spelling correction

Microsoft corporation provides word processor software like MS-Word, PowerPoint for spelling correction.

vi. Speech recognition

Speech recognition is used for converting spoken words in to text. It is used in applications, such as mobile, home automation, video recovery, dictating to Microsoft Word, voice biometrics, voice user interface, and so on.

vii. Chatbot

Implementing the Chatbot is one of the important applications of NLP. It is used by many companies to provide the customer's chat services.

viii. Information extraction

Information extraction is one of the most important applications of NLP. It is used for extracting structured information from unstructured or semi-structured machine-readable documents.